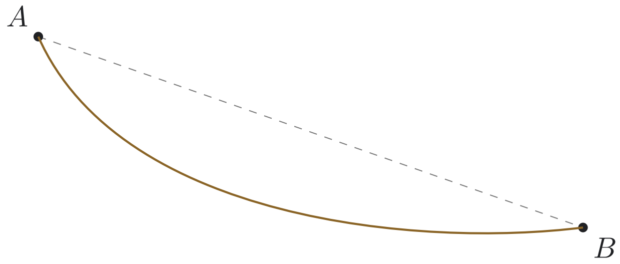
The Fastest Fall

Name: ____________________

Two fixed points, a frictionless bead, and a problem in which revealing structure matters more than naming the final curve.



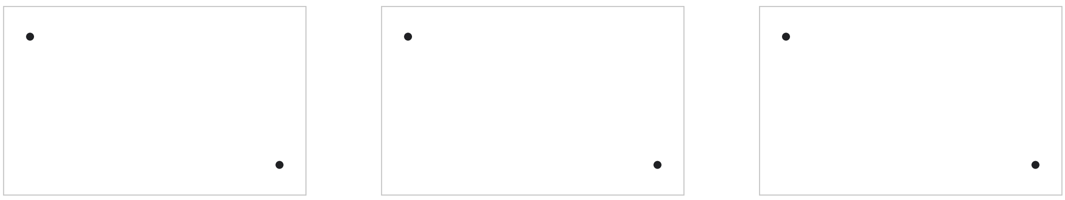
The question. A small bead is released from rest at A and slides without friction along a track to the lower point B. What shape of track gives the least travel time?

Working rule: You are not expected to solve the full optimisation problem. Progress includes a reasoned prediction, a calculation, a comparison, a model limitation, a useful representation, a counterexample, or a sharper question.

1. Predict before calculating

distance versus early speed

Sketch three possible tracks: a straight line, a deep dip, and a shallow curve. Rank them from fastest to slowest and justify your ranking.



Straight Deep dip Shallow

My initial ranking and reason:

2. Build a speed model

what can energy tell you?

At a vertical depth h below A, use conservation of energy to derive the bead's speed.

$$\frac{1}{2}mv^2 = mgh \qquad \Longrightarrow \qquad v = \underline{\hspace{4cm}}$$

State three consequences of this relation that are true before you know the track shape.

1. ____________________

2. ____________________

3. ____________________

3. Time as many small pieces — turn a curve into a calculation

For a short segment of length Δs travelled at approximately constant speed v,

$$\Delta t \approx \frac{\Delta s}{v}.$$

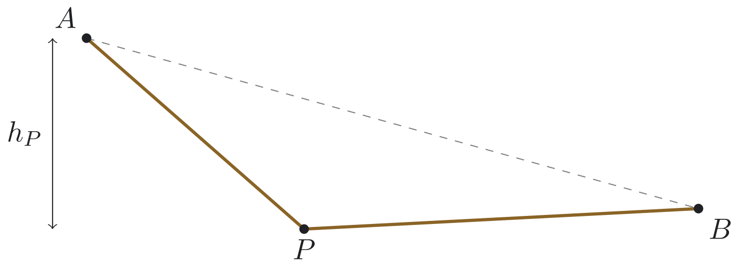
Complete a discrete model for the total time:

$$T \approx \sum_i \frac{\underline{\qquad\qquad}}{\sqrt{2g\,\underline{\qquad\qquad}}}.$$

Explain what each symbol represents and how you would estimate it from a track drawn on squared paper.

4. Compare a restricted family — two straight segments

Instead of searching over every possible curve, choose an intermediate point P and compare tracks $A \to P \to B$.



For a straight segment descending from depth h_1 to depth h_2, the speed varies along the segment. Describe one defensible approximation for its travel time. You may use an average speed, divide it into smaller pieces, or construct another method.

Record three choices of P and compare their estimated times.

Choice	Description of P	Estimated time	What this suggests
1			
2			
3			

5. Test the physical model — sliding, rolling, friction

A real marble rolls. For a solid sphere, $I = \frac{2}{5}mr^2$ and $\omega = v/r$.

Use

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

to show that

$$v = \sqrt{\frac{10gh}{7}}.$$

Does multiplying every speed by the same constant necessarily change which track is fastest? State a conjecture and support it.

List two further ways in which a real experiment would depart from the ideal model.

1. ______________________________

2. ______________________________

6. Use dimensions before solving — constrain the answer

Let H be a vertical scale, L a horizontal scale, and g the gravitational field strength. Show that a possible form for the minimum time is

$$T = \sqrt{\frac{H}{g}}\, f\!\left(\frac{L}{H}\right).$$

Explain why L/H must appear only inside a dimensionless function.

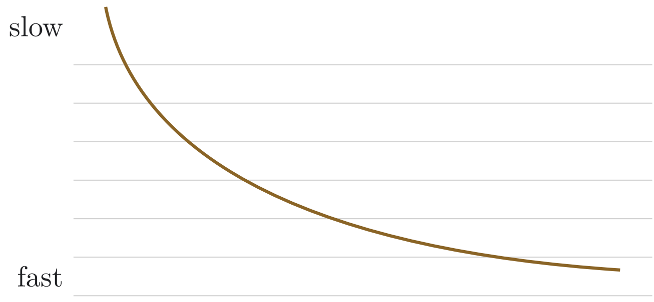
If every length is scaled by a factor λ, predict how the travel time scales:

$$T(\lambda) = \underline{\qquad\qquad\qquad} T(1).$$

Give a verbal explanation.

7. Translate the problem mechanics becomes refraction

Imagine dividing the vertical region into thin horizontal layers. Within each layer, treat the speed as constant.



Why might the track bend more strongly in the slow upper layers than in the fast lower layers?

A Snell-like relation can be written as

$$\frac{\sin\theta}{v} = \text{constant},$$

where θ is measured from the layer normal. Explain what this predicts as v increases with depth.

8. Push your conjecture to its limits limiting cases

Choose any three questions and respond without needing a complete proof.

- What if B is almost directly below A?
- What if B is very far horizontally from A?
- Could the best track dip below B and rise again?
- Should the track begin horizontally, vertically, or at some intermediate angle?
- Could an arbitrarily steep initial descent make the total time arbitrarily small?

9. Productive stuck — record the structure you have uncovered

Complete at least four of the following.

I can eliminate	A candidate track or claim that cannot be optimal, because...
I can identify an invariant	A quantity that does not affect result, or relation that stays fixed...
I can identify a scale	A combination of quantities that sets the characteristic time...
I can produce evidence	A calculation, measurement, graph, or numerical comparison that supports...
I can translate the problem	Another representation or area of physics that makes part of it clearer...
I can refine the model	An idealisation that should be replaced or qualified...
I can formulate the missing question	The next precise question whose answer would move the investigation forward...

10. Field notes — writing stabilises the model

Before discussion

I first assumed that...

This assumption became doubtful when...

One quantity I can calculate is...

A smaller version of the problem I could investigate is...

After discussion

Another person's idea changed my model because...

My current conjecture is...

The strongest evidence I have is...

The part I still cannot connect is...

11. Core results — what this enquiry actually earned

The relations established along the way.

$$v = \sqrt{2gh} \qquad T = \int_A^B \frac{ds}{v} \qquad T = \sqrt{\frac{H}{g}}\, f\left(\frac{L}{H}\right) \qquad T(\lambda) = \sqrt{\lambda}\, T(1)$$

No formula for the minimising curve itself appears above, and that omission is deliberate: every relation here was earned without it, which is the entire point of staging the problem this way.

Concepts earned, in order.

- Shortest distance and shortest time are different claims; a steep early descent trades extra path length for extra speed.
- Energy conservation gives the speed at any depth without needing to know the shape of the track that got you there.
- Dimensional analysis constrains every possible answer before any optimisation is attempted.
- Scaling every length by λ scales the time by $\sqrt{\lambda}$, a firm numerical fact obtained without solving anything.
- Treating speed as depth-dependent turns the mechanics problem into an optics problem, layer by layer.

For the record: the curve that actually solves this problem is called a *cycloid*. Its equation was deliberately withheld throughout, since the value of the investigation lay in establishing everything above it, not in being told the name.

Every substantial claim should be accompanied by a reason, calculation, measurement, diagram, simulation, or counterexample.